COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CpE31S2
DATE PERFORMED:	July 09, 2026	**DATE SUBMITTED:**	July 30, 2026
NAME:	Alvarez, Erick Jefferson A	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1

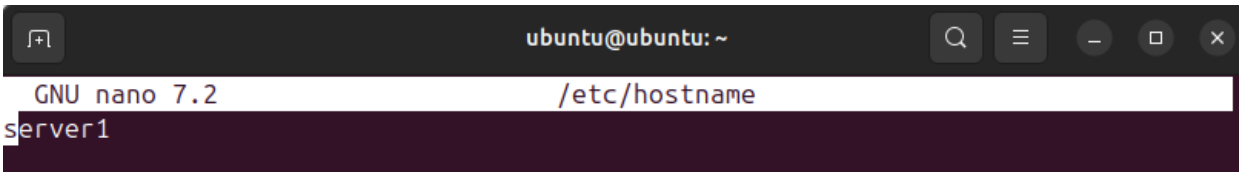
Creating Virtual Machines

1. PROCEDURE OUTPUT

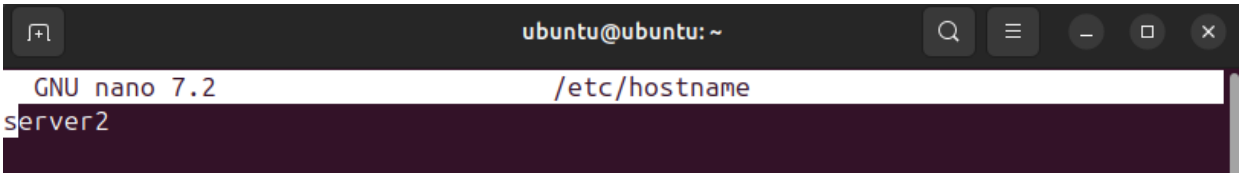
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

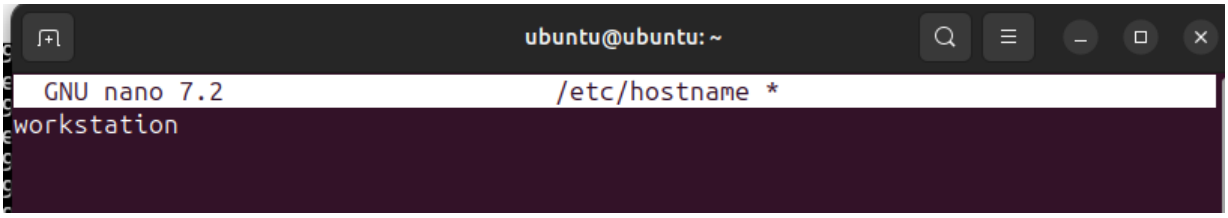
1.1 Use server1 for Server 1



1.2 Use server2 for Server 2

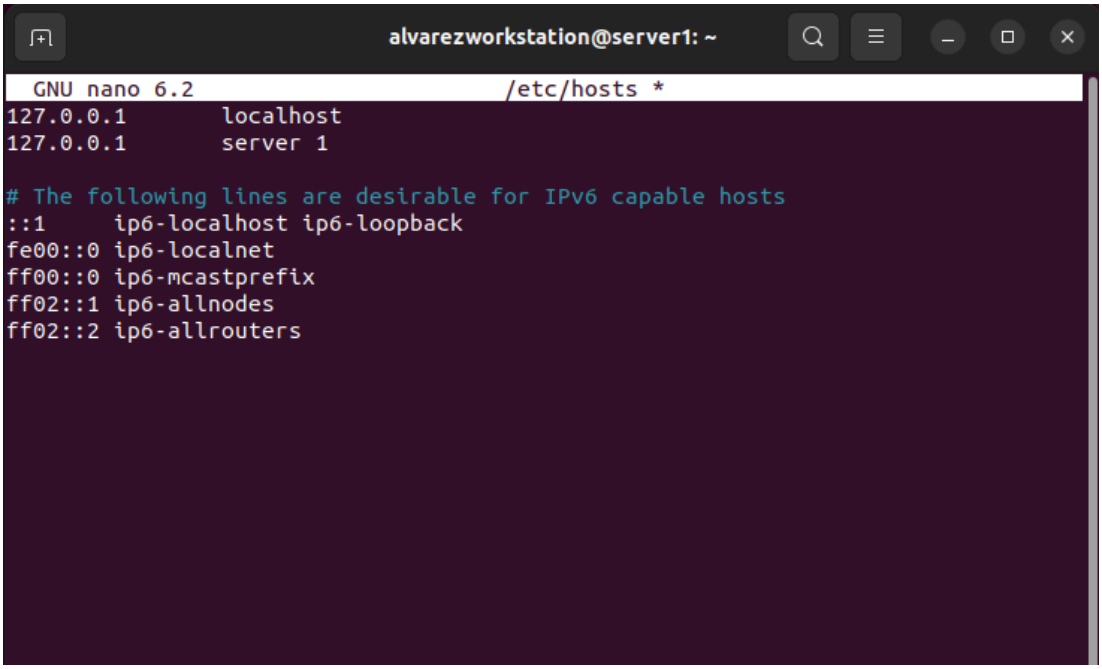


1.3 Use workstation for the Local Machine

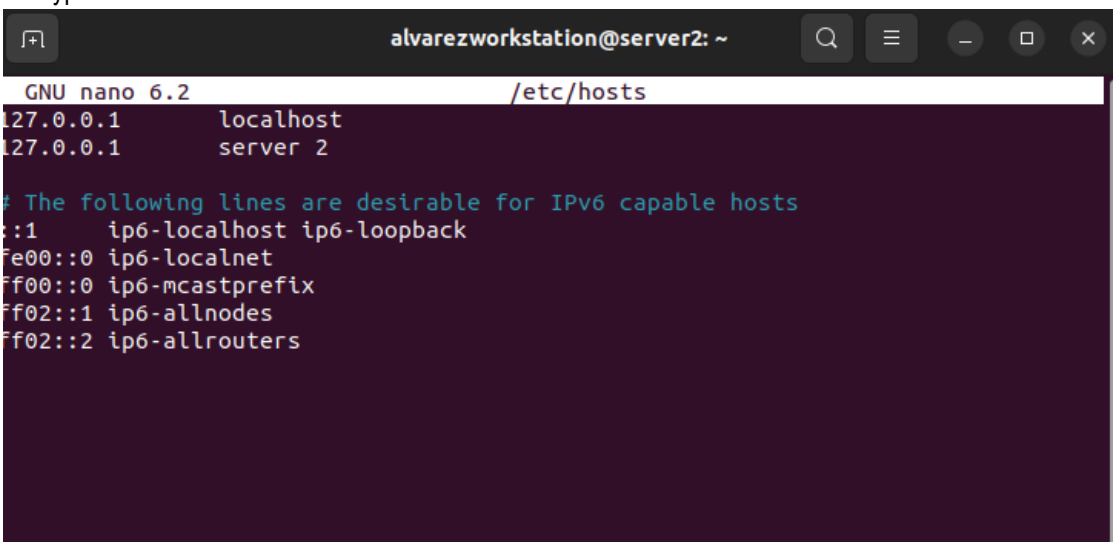


2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

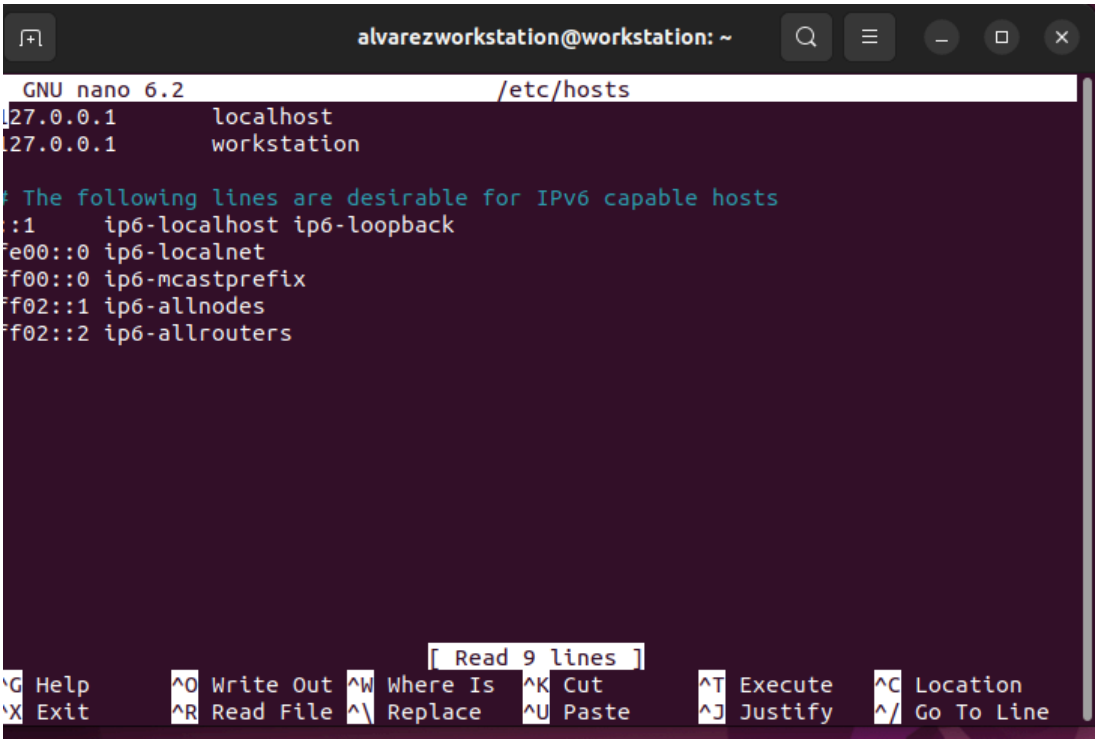
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine.



Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

Update:
server1:

```
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
 InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
 Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 k
B]
```

Server2:

```
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libfwupd3 libllvm20 linux-headers-6.17.0-35-generic
  linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17-tools-6.17.0-35
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic
  linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin
```

Workstation:

```
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
 InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
 Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 k
B]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827
kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 kB
```

Upgrade:

server1:

```
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libfwupd3 libllvm20 linux-headers-6.17.0-35-generic
  linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17-tools-6.17.0-35
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic
  linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin
The following packages have been kept back:
```

Server2:

```
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libfwupd3 libllvm20 linux-headers-6.17.0-35-generic
  linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17-tools-6.17.0-35
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic
  linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin
```

Workstation:

```
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35
  linux-hwe-6.17-tools-6.17.0-35 linux-image-6.17.0-35-generic
  linux-modules-6.17.0-35-generic linux-modules-extra-6.17.0-35-generic
  linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
```

2. Install the SSH server using the command sudo apt install openssh-server.

Workstation:

```
alvarezworkstation@workstation:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  openssh-server
```

Server 1:

```
alvarezworkstation@alvarezworkstation: ~
alvarezworkstation@alvarezworkstation:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
```

Server 2:

```
alvarezworkstation@alvarezworkstation: ~
alvarezworkstation@alvarezworkstation:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  openssh-server
```

3. Verify if the SSH service has started by issuing the following commands:
3.1 sudo service ssh start
Workstation:

```
alvarezworkstation@workstation:~$ sudo service ssh start
alvarezworkstation@workstation:~$
```

Server1:

```
alvarezworkstation@alvarezworkstation:~$ sudo service ssh start
alvarezworkstation@alvarezworkstation:~$
```

Server2:

```
alvarezworkstation@alvarezworkstation:~$ sudo service ssh start
alvarezworkstation@alvarezworkstation:~$
```

3.2 sudo systemctl status ssh

Workstation:

```
alvarezworkstation@workstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Thu 2026-07-16 10:01:59 CST; 5min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 4167 (sshd)
      Tasks: 1 (limit: 4597)
     Memory: 1.7M
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─4167 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 10:01:59 workstation systemd[1]: Starting OpenBSD Secure Shell server...
Jul 16 10:01:59 workstation sshd[4167]: Server listening on 0.0.0.0 port 22.
Jul 16 10:01:59 workstation systemd[1]: Started OpenBSD Secure Shell server.
Jul 16 10:01:59 workstation sshd[4167]: Server listening on :: port 22.
lines 1-16/16 (END)
```

Server1:

```
alvarezworkstation@alvarezworkstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Thu 2026-07-16 10:04:17 CST; 4min 14s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 4131 (sshd)
      Tasks: 1 (limit: 4597)
     Memory: 1.7M
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─4131 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 10:04:17 alvarezworkstation systemd[1]: Starting OpenBSD Secure Shell se>
Jul 16 10:04:17 alvarezworkstation sshd[4131]: Server listening on 0.0.0.0 port>
Jul 16 10:04:17 alvarezworkstation sshd[4131]: Server listening on :: port 22.
Jul 16 10:04:17 alvarezworkstation systemd[1]: Started OpenBSD Secure Shell ser>
lines 1-16/16 (END)
```

Server2:

```
alvarezworkstation@alvarezworkstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Thu 2026-07-16 10:06:01 CST; 3min 0s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 4176 (sshd)
      Tasks: 1 (limit: 4597)
     Memory: 1.7M
        CPU: 13ms
     CGroup: /system.slice/ssh.service
             └─4176 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 10:06:01 alvarezworkstation systemd[1]: Starting OpenBSD Secure Shell se>
Jul 16 10:06:01 alvarezworkstation sshd[4176]: Server listening on 0.0.0.0 port>
Jul 16 10:06:01 alvarezworkstation sshd[4176]: Server listening on :: port 22.
Jul 16 10:06:01 alvarezworkstation systemd[1]: Started OpenBSD Secure Shell ser>
lines 1-16/16 (END)
```

4. Configure the firewall to all port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
Workstation:

```
alvarezworkstation@workstation: ~
alvarezworkstation@workstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
alvarezworkstation@workstation:~$
```

Server1:

```
alvarezworkstation@alvarezworkstation: ~
alvarezworkstation@alvarezworkstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

Server2:

```
alvarezworkstation@alvarezworkstation: ~
alvarezworkstation@alvarezworkstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

4.2 sudo ufw enable
Workstation:

```
alvarezworkstation@workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

Server1:

```
alvarezworkstation@alvarezworkstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

Server2:

```
alvarezworkstation@alvarezworkstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
alvarezworkstation@alvarezworkstation:~$
```

4.3 sudo ufw status

Workstation:

```
alvarezworkstation@workstation:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

alvarezworkstation@workstation:~$
```

Server1:

```
alvarezworkstation@alvarezworkstation:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

alvarezworkstation@alvarezworkstation:~$
```

Server2:

```
alvarezworkstation@alvarezworkstation:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

alvarezworkstation@alvarezworkstation:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.120

```
192.168.56.120
```

1.2 Server 2 IP address: 192.168.56.121

```
192.168.56.121
```

1.3 Local Machine IP address: 192.168.56.119

```
192.168.56.119
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☑ Successful ☐ Not Successful

```
alvarezworkstation@workstation:~$ ping 192.168.56.120
PING 192.168.56.120 (192.168.56.120) 56(84) bytes of data.
64 bytes from 192.168.56.120: icmp_seq=1 ttl=64 time=1.49 ms
64 bytes from 192.168.56.120: icmp_seq=2 ttl=64 time=2.71 ms
64 bytes from 192.168.56.120: icmp_seq=3 ttl=64 time=1.70 ms
64 bytes from 192.168.56.120: icmp_seq=4 ttl=64 time=0.586 ms

--- 192.168.56.120 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 9070ms
rtt min/avg/max/mdev = 0.586/1.621/2.708/0.754 ms
^Calvarezworkstation@workstation:~$
```

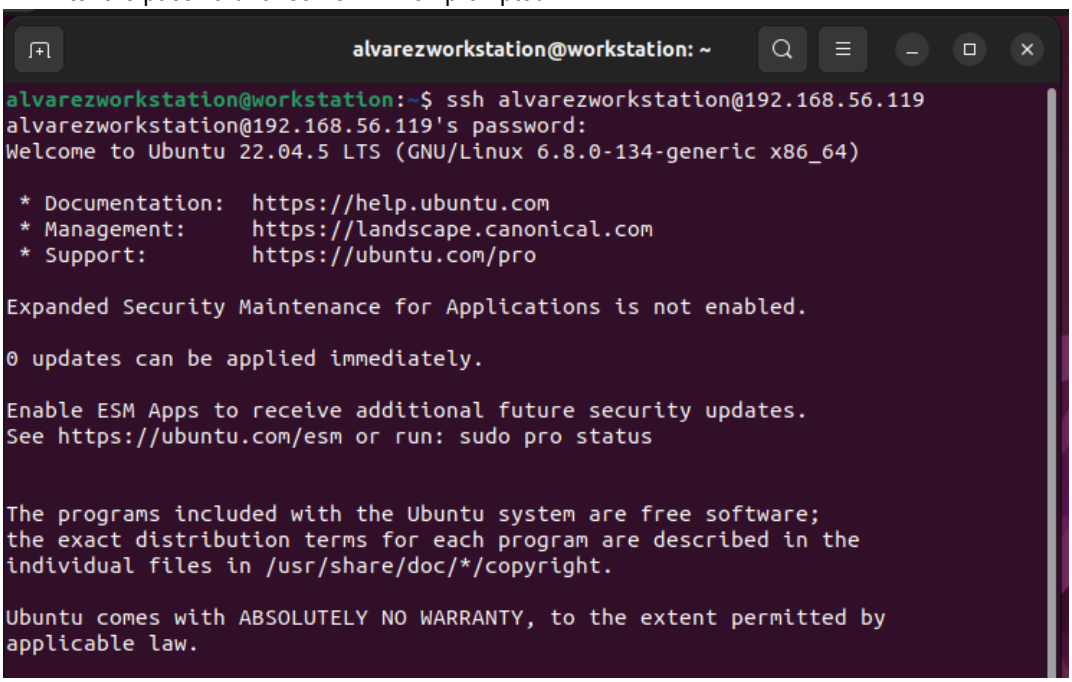
2.2 Connectivity test for Local Machine 1 to Server 2: ☑ Successful ☐ Not Successful

```
alvarezworkstation@workstation:~$ ping 192.168.56.121
PING 192.168.56.121 (192.168.56.121) 56(84) bytes of data.
64 bytes from 192.168.56.121: icmp_seq=1 ttl=64 time=0.631 ms
64 bytes from 192.168.56.121: icmp_seq=2 ttl=64 time=3.26 ms
64 bytes from 192.168.56.121: icmp_seq=3 ttl=64 time=0.941 ms
64 bytes from 192.168.56.121: icmp_seq=4 ttl=64 time=1.04 ms
^C
--- 192.168.56.121 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 7714ms
rtt min/avg/max/mdev = 0.631/1.468/3.264/1.047 ms
alvarezworkstation@workstation:~$
```

2.3 Connectivity test for Server 1 to Server 2: ☑ Successful ☐ Not Successful

```
alvarezworkstation@server1:~$ ping 192.168.56.121
PING 192.168.56.121 (192.168.56.121) 56(84) bytes of data.
64 bytes from 192.168.56.121: icmp_seq=1 ttl=64 time=1.90 ms
64 bytes from 192.168.56.121: icmp_seq=2 ttl=64 time=0.347 ms
64 bytes from 192.168.56.121: icmp_seq=3 ttl=64 time=0.541 ms
64 bytes from 192.168.56.121: icmp_seq=4 ttl=64 time=0.484 ms
^C
--- 192.168.56.121 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.347/0.818/1.901/0.629 ms
alvarezworkstation@server1:~$
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.
1. On the Local Machine, issue the following commands:
1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120
1.2 Enter the password for server 1 when prompted



1.3 Verify that you are in server 1. The user should be in this format user@server1.
For example, jvtaylar@server1

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

alvarezworkstation@server1:~$
```

2. Logout of Server 1 by issuing the command control + D.

```
alvarezworkstation@server1:~$
logout
Connection to 192.168.56.120 closed.
alvarezworkstation@server1:~$
```

3. Do the same for Server 2.

```
alvarezworkstation@server2:~$ ssh alvarezworkstation@192.168.56.121
The authenticity of host '192.168.56.121 (192.168.56.121)' can't be establish
ED25519 key fingerprint is SHA256:gpnvZ4AQxgwSPr2+FY6OObdVCc3zERcDmOu0tCHcUoQ
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.121' (ED25519) to the list of known ho
.
alvarezworkstation@192.168.56.121's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```

4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

```
alvarezworkstation@workstation: ~
  GNU nano 6.2                          /etc/hosts
127.0.0.1       localhost
127.0.0.1       workstation
192.168.56.120  server1
192.168.56.121  server2

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
```

4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Server 1:

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 08:08:09 2026 from 192.168.56.120
alvarezworkstation@server1:~$
logout
Connection to 192.168.56.120 closed.
alvarezworkstation@workstation:~$
```

Server 2:

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 08:09:46 2026 from 192.168.56.121
alvarezworkstation@server2:~$
logout
Connection to 192.168.56.121 closed.
alvarezworkstation@workstation:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - Hostname is easier than IP address in SSH because the DNS modifies the hostname into the correct IP address. This method makes it easier to remember.

2. How secured is SSH?
   - The SSH is very secure because it makes sure that the connection between your computer and the server is encrypted. This encryption makes it harder for others to access your information. SSH can also confirm whether the server you're logging into is the correct one you want to connect to.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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